MIT 8.012 Problem Set 4 - Problem 3.20 (K&K)

3.20 A rocket ascends from rest in a uniform gravitational field by ejecting exhaust with constant speed u. Assume that the rate at which mass is expelled is given by $dm/dt = \gamma m$, where m is the instantaneous mass of the rocket and γ is a constant, and that the rocket is retarded by air resistance with a force mbv, where b is a constant. Find the velocity of the rocket as a function of time.

Solution (handwritten) follows on the next page.

Problem 8

By the Rocket Equation

$$F_{ext} = m\frac{dv}{dt} - u\frac{dm}{dt}$$

$$m(g - bv) = m\left(\frac{dv}{dt} + uv\right)$$

$$\frac{dv}{dt} = g - uv - bv$$

$$-\frac{\ln(g - uv - bv)}{b} = t + c$$

$$g - uv - bv = -c \cdot e^{-bt}$$

$$-bv = c \cdot e^{-bt} + uv - g$$

$$v = \frac{-c \cdot e^{-bt} - uv + g}{b}$$

$$v(0) = 0$$

$$-c = -g + uv \Rightarrow c = -uv + g$$

$$v = \frac{(g - uv)(1 - e^{-bt})}{b}$$